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Quantitative Analysis of Jane Austen's Works with Voyant

As Moretti claims, the growing expanse of literature has induced the need for distant reading methods to properly analyze the network. In this lab, we use these computational methods to analyze a particular subgraph of the overall structure by focusing on the works of Jane Austen. In particular, we analyzed 9 novels by Jane Austen and her publicly available personal letters from the Project Gutenberg repository using the digital humanities tool Voyant. Chosen to capture longitudinal trends, this corpus spans Austen's career starting with *Love and Freindship* written in 1790 and concluding with posthumously published books such as *Persuasion*. Finally, we included Austen's personal letters to analyze her writing in another dimension orthogonal to her novels and explore possible similarities to her epistolary works. Following the guidance of Schofield, we started by preprocessing the dataset to remove boilerplate from the Project Gutenberg website for which we were fortunate to use libraries from others who had encountered this before (Kaser and Lemier). This data cleaning step removed hundreds of occurrences of words such as "Gutenberg", "Project", and "ebook" which would have resulted in erroneous analysis. We also extended the stop words list by adding the words "mr", "mrs" and "miss" to further focus our analysis. This refinement is displayed in Figure 1 as the most common words in the dataset start to carry more meaning for our study.

Figure 1: Successive refinement of the corpus by removing
Project Gutenberg headers and extending stop words

We started analysis using the terms pane to extend our close reading intuitions and try to extrapolate towards larger trends within the corpus. Inspired by our previous knowledge of the complex interactions between marriage and wealth in *Pride and Prejudice*, we centered our analysis on these themes. To do so, we first began by operationalizing each theme with a group of words. For example, to analyze the concept of marriage, we used the processes of lemmatization to create the set “marriage*”, “marry*”, “wife*”, “husband*” and “proposal*”. This method allowed us to combine similar forms of words and more broadly capture the notion of marriage. We then analyzed the relative frequencies of each of these across the entire corpus. By selecting relative frequencies, we were able to normalize for the varied lengths of works in the corpus and standardize our results.

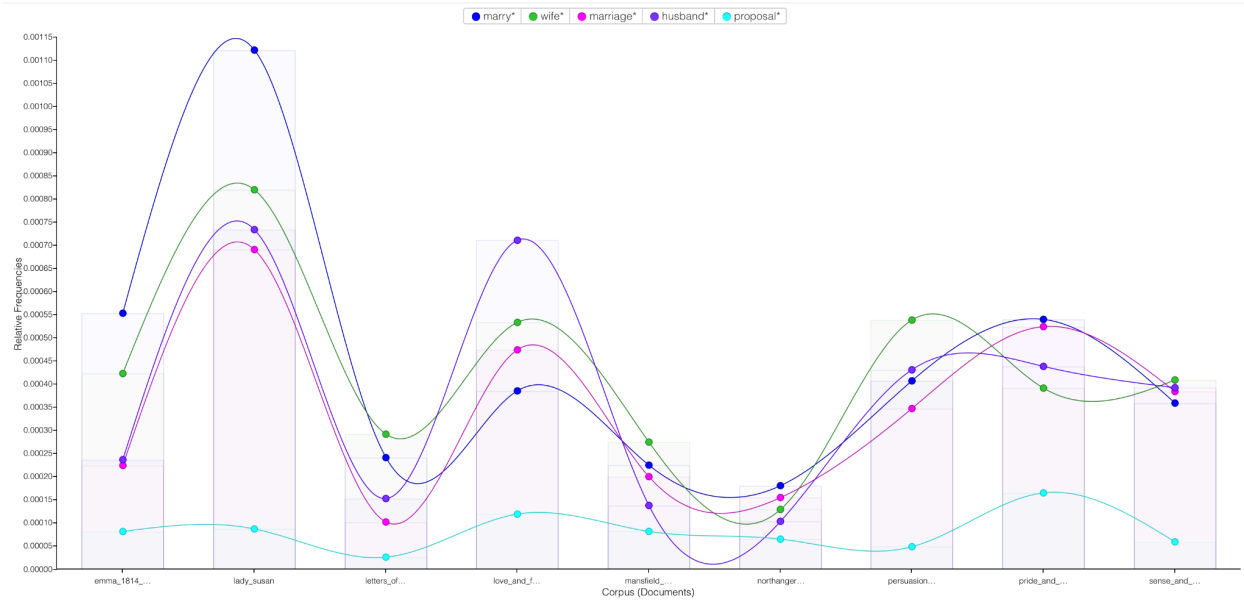

Figure 2: Relative Frequency distribution of words associated with the theme of marriage

As seen in Figure 2, these words followed the hypothesized covariance we expected to see by having similar trends for relative frequency. Some works stand out as interacting with these concepts particularly strongly such as *Lady Susan* and *Love and Freinds* whereas others contain these concepts relatively infrequently such as *Northanger Abbey* and *Mansfield Park*. Contextualizing these works throughout Austen's life, we see that earlier works tended to contain more occurrences of the theme of marriage, but the concept appeared less frequently through her middle novels before finally returning again in her final works. However, the data still needs to be contextualized and interpreted before claims can be made about the importance of marriage across the various novels. It could be the case that these concepts are simply less prominent in certain works not because marriage is thematically absent, but because Austen engages with the institution through implication or irony rather than direct naming. This distinction illustrates exactly the kind of interpretive limitation Schofield warns against: high frequency alone does not confirm thematic centrality, and low frequency does not preclude it. As an example, we see in Figure 3 that the while the word love does agree with the general trend seen by the concept of

marriage in some works, it does not for others. In particular, though *Lady Susan* had an extremely high, in fact the highest, relative frequency for words similar to marriage, it did not have the same high relative frequency for the word “love”.

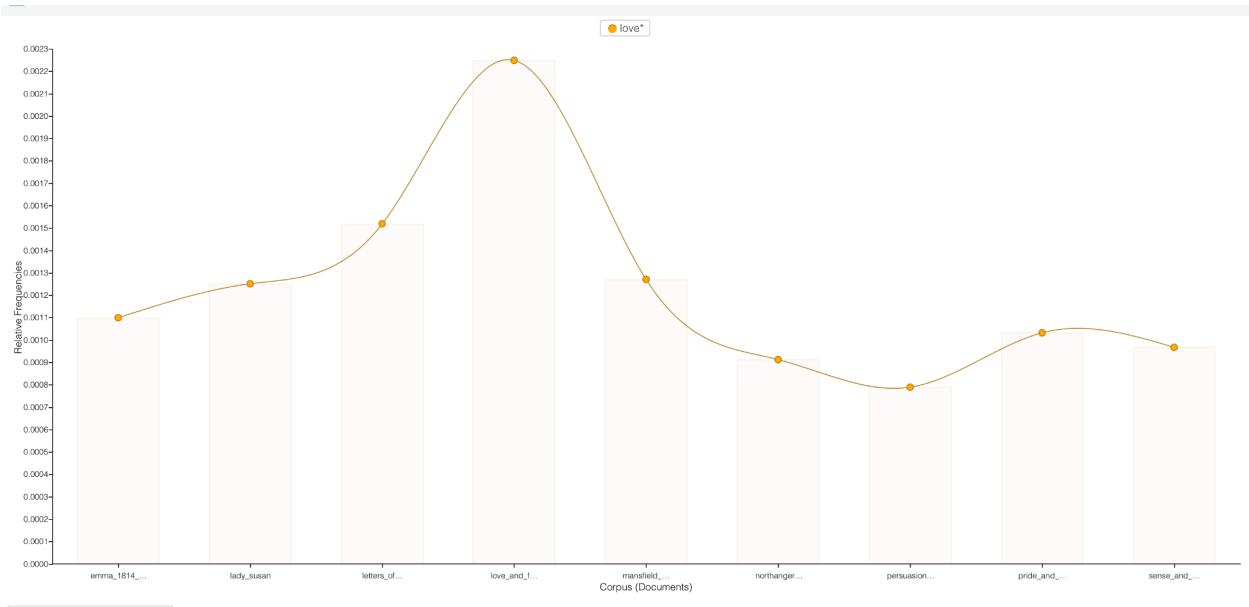

Figure 3: Relative Frequency distribution of “love”

These limitations were further seen when we analyzed the theme of wealth. Again, we operationalized this concept with a term set. We selected “money*”, “pounds*”, “income*”, “estate*” and “property*”. Unlike our previous findings, the correlations between the different words in this set are unpredictable and don’t show clear trends through the corpus.

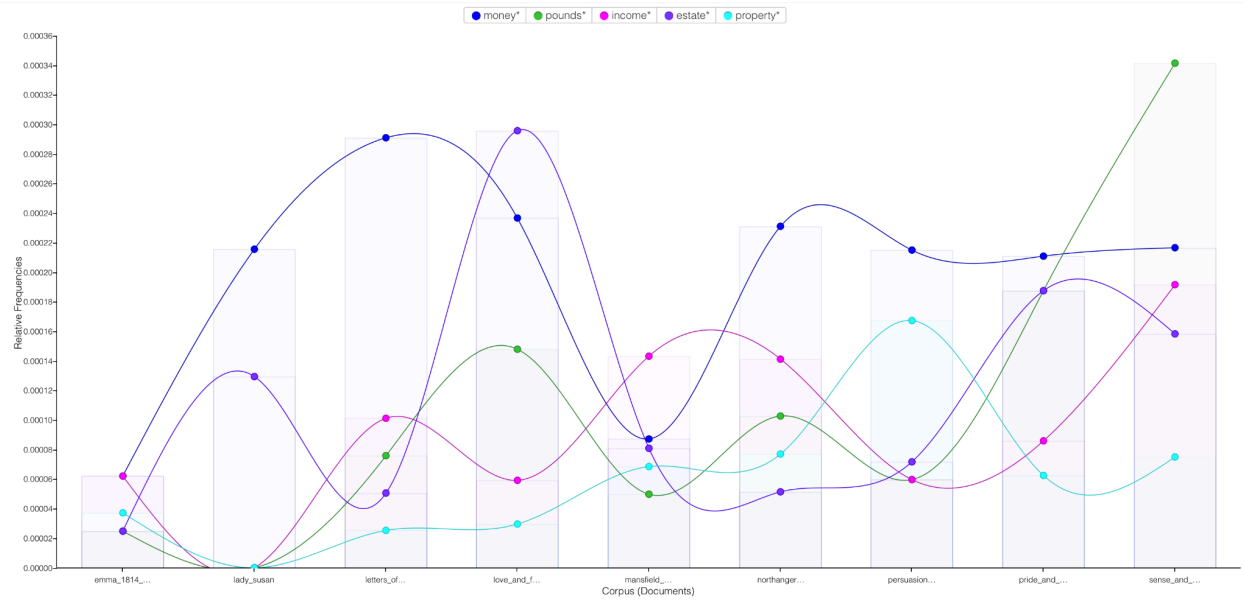

Figure 4: Relative Frequency distribution of words associated with the theme of wealth

However, there are still some quantitative results we can analyze. For example, the concept of property seems to be more emphasized in Austen's later works, possibly reflecting Austen's deepened consideration of economic realities during the period. Furthermore, we also see the extremely high relative frequency of Austen's use of the word "money" in her personal letters which contrasts sharply from the previous low relative frequency of words relating to marriage. Examining the nearby context of the word "money", we see that Austen is using the word in a fairly pragmatic way in her personal correspondence. There is little of the ironic distance or narrative framing that characterizes its use in the novels. This contrast is itself a meaningful finding. Whereas Austen's novels aestheticize financial precarity and transform it into the plot, the letters address the topic directly.

This disparity further emphasizes the importance of curating a diverse corpus to allow for various modes of analysis. By including Austen's personal letters, we discover the differences in Austen's styles between various settings. Of course, these results were not simply found by quantitative analysis, but rather by the questions it induced to fill the gaps. This analysis could be taken further with close reading if the data was more structured. With simple text files, much of the media specific analysis, whose importance is highlighted by Hayles, is impossible to perform. This could be especially important in the context of the personal letters and the comparison to Austen's novels.

Though our quantitative analysis did highlight findings about trends and themes in Austen's works, it is important to reflect on the many choices we made along the way. By selecting a corpus, specific operational definitions, and methods of analysis, we are at risk of introducing bias into our analysis. However, by following the practices encouraged by Drucker and Schofield, we reduce our susceptibility to these pitfalls by consciously reflecting upon our method of analysis and our use of these quantitative tools.

Works Cited

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